Akoizo Original Practice - Form 2 - Answer key

Division B | Crime Busters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which listed metal does not react with hydrochloric acid?

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

A. Magnesium

B. Iron

C. Copper

D. Zinc

Correct: C

2. (2 points) Which plastics float in water?

Unit: Recyclable plastics.
Worked-case data: A plastic fragment sinks in water, floats in a 1.10 g/cm^3 salt solution, and burns with a sooty flame. Identify it.
Separate investigation observations:
Water: Sinks.
1.10 g/cm^3 solution: Sinks.
1.25 g/cm^3 solution: Floats.
Appearance: Very clear, rigid, and shatter-resistant.

A. PVC and PETE

B. PS and PMMA

C. All plastics

D. PP, LDPE, and HDPE

Correct: D

3. (2 points) Which fingerprint pattern has no delta?

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

A. Arch

B. Loop

C. Whorl

D. Double loop

Correct: A

4. (2 points) Which solid fizzes in plain water without added acid?

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

A. Baking soda

B. Calcium carbonate

C. Sugar

D. Alka-Seltzer

Correct: D

5. (2 points) A fiber burns quickly, glows afterward, and leaves soft gray ash. It is:

A. Vegetable (such as cotton)

B. Animal (such as wool)

C. Synthetic (such as nylon)

D. Mineral

Correct: A

6. (2 points) A spot travels 2.0 cm while the solvent travels 8.0 cm. What is its Rf?

A. 2.5

B. 0.25

C. 4.0

D. 0.40

Correct: B

7. (2 points) Which evidence should usually carry the most weight?

Unit: Writing the crime scene analysis.
Worked-case data: Evidence: the scene powder is baking soda; suspect A has baking soda, B has cornstarch, C has baking soda. Note ink matches only C's pen. A dog hair was found; A and C own dogs, B owns a cat. Who is the culprit?
Separate investigation observations:
Powder: Calcium carbonate: found with suspects 1 and 3.
Plastic: HDPE fragment: suspects 1, 2, and 3 all use HDPE jugs.
Chromatography: Note ink matches only suspect 3's pen.
Fingerprint: Scene loop matches suspect 3's minutiae.

A. Unknown odors

B. Suspect statements

C. Individual evidence, such as a full fingerprint

D. Common class evidence

Correct: C

8. (2 points) Adding yeast to which liquid causes vigorous bubbling?

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

A. Vinegar

B. Hydrogen peroxide

C. Water

D. Isopropyl alcohol

Correct: B

9. (2 points) What is resin code 2?

A. PVC

B. LDPE

C. HDPE

D. PETE

Correct: C

10. (2 points) A DNA profile matches a suspect only if:

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

A. Most peaks match

B. One location matches

C. The peaks are the same height

D. Every peak at every location matches

Correct: D

11. (4 points) Describe how to distinguish salt from sugar without tasting.

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

answer: Under a lens, salt forms clear cubic crystals while sugar crystals are irregular and slightly elongated. Both dissolve, so crystal shape is the key observation. (4 points).

12. (4 points) Describe the three layers of a hair.

Unit: Hair and fibers.
Worked-case data: A hair has a continuous medulla about 60% of the hair's width and scales shaped like petals. What is its source?
Separate investigation observations:
Hair A: Medulla fragmented; index about 0.2.
Hair B: Wide, continuous ladder-like medulla; flattened scales.
Fiber: Burns quickly with afterglow and soft gray ash.

answer: The cuticle is the scaly outer layer, the cortex holds pigment, and the medulla is the central canal. (4 points).

13. (4 points) Why must known and unknown samples be run on the same paper at the same time?

answer: Rf values depend on solvent, paper, and conditions, so only side-by-side runs are directly comparable. (4 points).

14. (4 points) Why is a summary table helpful before writing?

Unit: Writing the crime scene analysis.
Worked-case data: Evidence: the scene powder is baking soda; suspect A has baking soda, B has cornstarch, C has baking soda. Note ink matches only C's pen. A dog hair was found; A and C own dogs, B owns a cat. Who is the culprit?
Separate investigation observations:
Powder: Calcium carbonate: found with suspects 1 and 3.
Plastic: HDPE fragment: suspects 1, 2, and 3 all use HDPE jugs.
Chromatography: Note ink matches only suspect 3's pen.
Fingerprint: Scene loop matches suspect 3's minutiae.

answer: It shows which evidence links or excludes each suspect, revealing the culprit and keeping the explanation organized. (4 points).

15. (4 points) Describe how to tell vinegar from lemon juice.

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

answer: Both are acidic, so use odor and appearance: vinegar is clear with a sharp sour smell, while lemon juice is cloudy and yellowish with a citrus smell. (4 points).

16. (4 points) How could you separate HDPE from PP using density?

Unit: Recyclable plastics.
Worked-case data: A plastic fragment sinks in water, floats in a 1.10 g/cm^3 salt solution, and burns with a sooty flame. Identify it.
Separate investigation observations:
Water: Sinks.
1.10 g/cm^3 solution: Sinks.
1.25 g/cm^3 solution: Floats.
Appearance: Very clear, rigid, and shatter-resistant.

answer: Use a solution with density between them, about 0.92-0.94 g/cm^3. PP (0.90) floats and HDPE (0.95) sinks. (4 points).

17. (4 points) How can tire tracks show direction of travel?

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

answer: Soil or debris is thrown backward by spinning tires, and tread marks can show the direction the vehicle moved. (4 points).

18. (4 points) How would you separate a mixture of salt and sand?

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

answer: Add water to dissolve the salt, filter or decant to remove the sand, then evaporate the liquid to recover the salt. (4 points).